Exercice 1

1/ $$\underline{s} = \frac{R}{jL\omega + r + \frac{1}{jC\omega} + R}\,\underline{e}$$

$$\rightarrow \boxed{\underline{H}_1 = \frac{\underline{s}}{\underline{e}} = \frac{jRC\omega}{1 + j(r+R)C\omega - \omega^2 LC}} \quad (1)$$

2/ $$\boxed{\underline{H}_2 = \frac{\underline{e}}{\underline{s}} = 1 + \frac{R_2}{R_1}} \quad (2)$$

3/ $$\underline{H}_1 \times \underline{H}_2 = \frac{\underline{s}}{\underline{e}} \times \frac{\underline{e}}{\underline{s}} = 1$$

$$= \left(1 + \frac{R_2}{R_1}\right) \frac{jRC\omega}{1 + j(r+R)C\omega - LC\omega^2}$$

$$\left(1 + \frac{R_2}{R_1}\right) jCR\omega = 1 + j(r+R)C\omega - LC\omega^2$$

$$\rightarrow \quad C\left(r + \cancel{R} - \cancel{R} - \frac{RR_2}{R_1}\right) j\omega + 1 - LC\omega^2 = 0$$

$$\rightarrow \begin{cases} 1 - LC\omega^2 = 0 \\ C\left(r + \quad - \frac{R_2 R}{R_1}\right) = 0 \end{cases} \rightarrow \boxed{\begin{cases} \omega = \frac{1}{\sqrt{LC}} \\ r\,R_1 = R\,R_2 \end{cases}}$$

4/ L'amplitude de e est bornée par la saturation → 15 V

or $e = \frac{R_1}{R_1+R_2} s$ → amplitude de s $= \frac{R_1+R_2}{R_1} \times 15V$

5/ C'est $\underline{s}$ en sortie du passe-bande

6/
(1) → $\left(1 + j(r+R)C\omega - \omega^2 LC\right)\underline{s} = (jRC\omega)\underline{e}$

or (2) → $\underline{e} = \left(1 + \frac{R_2}{R_1}\right)\underline{s}$

↳ $\left(1 + j(r+R)C\omega - \omega^2 LC\right)\underline{s} = jRC\omega\left(1+\frac{R_2}{R_1}\right)\underline{s}$

→ $\underline{s} + \left(\cancel{RC} + rC - \cancel{RC} - \frac{RR_2}{R_1}C\right) j\omega \underline{s}$

$+ LC(j\omega)^2 \underline{s} = 0$

→ $s(t) + \left(rC - \frac{RR_2}{R_1}C\right)\frac{ds}{dt} + LC\frac{d^2s}{dt^2} = 0$

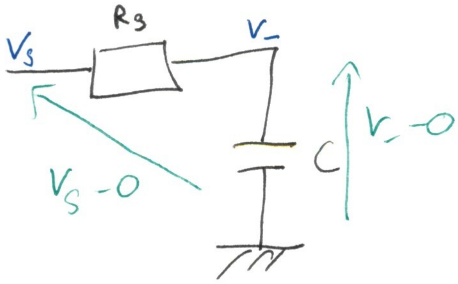
→ Les oscillations démarrent ssi le système est instable ssi

$$rC - \frac{R R_2}{R_1} C < 0$$

$$\Leftrightarrow \boxed{r < \frac{R R_2}{R_1}}$$

Exercice 2

1/



filtre RC série

$$\text{pdt : } \underline{V_-} - 0 = \frac{1/jC\omega}{\frac{1}{jC\omega} + R_3}(\underline{V_s} - 0)$$

$$\rightarrow \underline{V_-} = \frac{1}{1 + jR_3C\omega}\underline{V_s}$$

$$\rightarrow \boxed{\underline{H}_{RC} = \frac{\underline{V_-}}{\underline{V_s}} = \frac{1}{1 + jR_3C\omega}}$$

$$\underline{V_-}(1 + jR_3C\omega) = \underline{V_s}$$

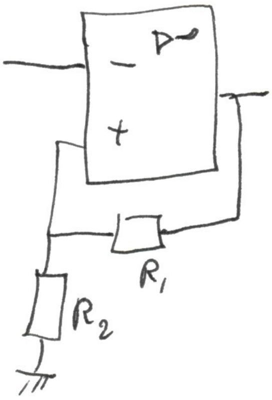
$$\underline{V_-} + R_3C\,j\omega\,\underline{V_-} = \underline{V_s}$$

$$V_-(t) + R_3C\frac{dV_-}{dt} = V_s(t)$$

2/ $V_-(t) = SGEH(t) + SPEASM.$

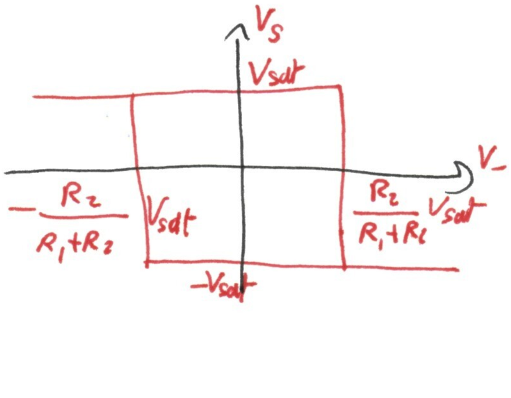
$$\boxed{= \lambda e^{-\frac{t}{R_3C}} + V_s}$$

3/



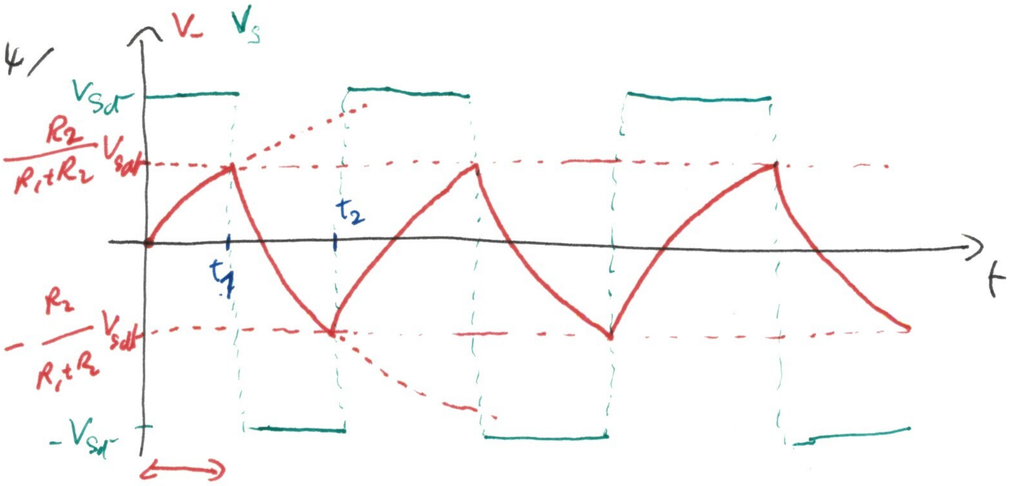
est un C.H.N.

Sa caractéristique est



⚠ Les noms de R_1 et R_2 sont inversés par rapport au cours.

4/



⚠ Pas $\frac{T}{2}$ (ni $\frac{T}{4}$)

5/ $T = t_2 - t_1$

entre t_1 et t_2 $V_s = -V_{sat}$

$$V_-(t) = \lambda e^{-t/R_3C} - V_{sat}$$

or à $t = t_1$, $V_-(t_1) = \frac{R_2}{R_1+R_2}V_{sat} = \lambda e^{-t_1/R_3C} - V_{sat}$ (1)

et à $t = t_2$ $V_-(t_2) = -\frac{R_2}{R_1+R_2}V_{sat} = \lambda e^{-t_2/R_3C} - V_{sat}$ (2)

$$\frac{(1)+V_{sat}}{(2)+V_{sat}} : \quad \frac{\frac{R_2}{R_1+R_2}\cancel{V_{sat}} + \cancel{V_{sat}}}{-\frac{R_2}{R_1+R_2}\cancel{V_{sat}} + \cancel{V_{sat}}} = \frac{\lambda e^{-t_1/R_3C} \cancel{-V_{sat}} \cancel{+V_{sat}}}{\lambda e^{-t_2/R_3C} \cancel{-V_{sat}} \cancel{+V_{sat}}}$$

$$\frac{\frac{R_2+R_1+R_2}{\cancel{R_1+R_2}}}{\frac{\cancel{-R_2}+R_1+\cancel{R_2}}{\cancel{R_1+R_2}}} = \frac{e^{-t_1/R_3C}}{e^{-t_2/R_3C}} = e^{\frac{t_2-t_1}{R_3C}}$$

$$\frac{2R_2+R_1}{R_1} = e^{\frac{t_2-t_1}{R_3C}}$$

$$\rightarrow \ln\left(1+2\frac{R_2}{R_1}\right) = \frac{t_2-t_1}{R_3C}$$

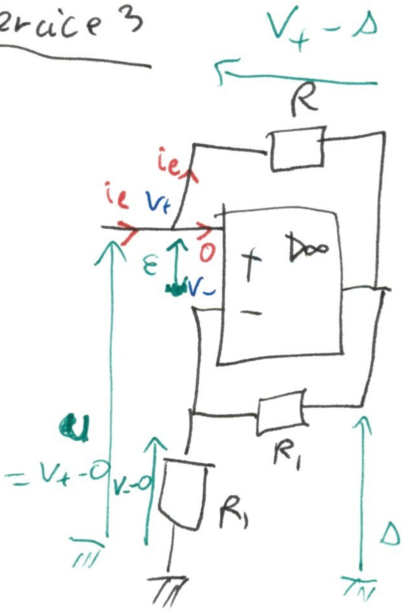
$$\rightarrow t_2 - t_1 = R_3C\ln\left(1+2\frac{R_2}{R_1}\right)$$

or $t_2 - t_1 = \frac{T}{2}$

$$\rightarrow \boxed{T = 2 R_3 C \ln\left(1 + 2\frac{R_2}{R_1}\right)}$$

Exercice 3

1/



$\begin{cases} \text{ALI idéal} \\ \text{stable} \end{cases} \rightarrow \varepsilon = 0 \rightarrow V_+ = V_-$

or $u = V_+ - 0 = V_+$

$\rightarrow V_- = u$

pdt : $V_- = \dfrac{R_1}{R_1 + R_1} s = \dfrac{1}{2} s$

$\rightarrow \boxed{u = \dfrac{1}{2} s}$

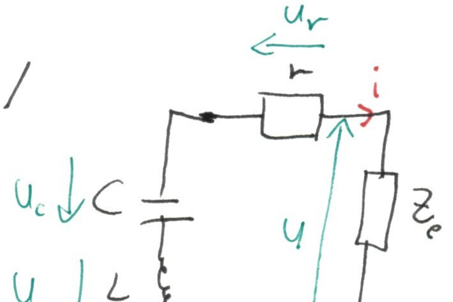
2/

$$i_e = \frac{V_+ - s}{R} = \frac{u - s}{R} = \frac{u - 2u}{R} = -\frac{u}{R}$$

$\rightarrow u = -R i_e$

$\rightarrow \boxed{Z_e = \dfrac{u}{i_e} = -R}$

3/



Loi des mailles : $u + u_r + u_c + u_L = 0$

$\rightarrow Z_e i + r i + u_c + L \dfrac{di}{dt} = 0$

$\rightarrow (Z_e + r) \dfrac{di}{dt} + \underbrace{\dfrac{du_c}{dt}}_{\frac{1}{C} i} + L \dfrac{d^2 i}{dt^2} = 0$

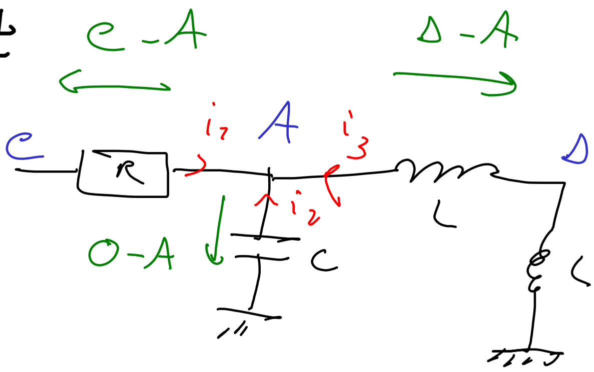
$\rightarrow \boxed{\dfrac{d^2 i}{dt^2} + \dfrac{r - R}{L} \dfrac{di}{dt} + \dfrac{1}{LC} i = 0}$

Pour que les oscillations démarrent, le système doit être instable $\rightarrow r - R < 0$

$\rightarrow r < R$

Exercice 4

1/



$$i_1 + i_2 + i_3 = 0$$

$$\frac{\underline{e} - \underline{A}}{R} + \frac{0 - \underline{A}}{1/jC\omega} + \frac{\underline{D} - \underline{A}}{jL\omega} = 0$$

$$\underline{A}\left[\frac{1}{R} + jC\omega + \frac{1}{jL\omega}\right] = \frac{\underline{e}}{R} + \frac{\underline{D}}{jL\omega}$$

$$\boxed{\underline{A} = \frac{\underline{e}/R + \underline{D}/jL\omega}{\frac{1}{R} + \frac{1}{jL\omega} + jC\omega}}$$

2/ $$\underline{D} = \underline{A}\,\frac{jL\omega}{jL\omega + jL\omega} = \frac{\underline{A}}{2}$$

$$\rightarrow \quad \underline{A} = 2\underline{D}$$

$$2\underline{D} = \frac{\underline{e}/R + \underline{D}/jL\omega}{\frac{1}{R} + \frac{1}{jL\omega} + jC\omega}$$

$$\to \underline{s}\left[2 - \frac{1/jL\omega}{\frac{1}{R}+\frac{1}{jL\omega}+jC\omega}\right]$$

$$= \underline{e}\,\frac{1/R}{\frac{1}{R}+\frac{1}{jL\omega}+jC\omega}$$

$$= \frac{\frac{2}{R}+\frac{2}{jL\omega}+2jC\omega-\frac{1}{jL\omega}}{\frac{1}{R}+\frac{1}{jL\omega}+jC\omega}$$

$$\boxed{\underline{H} = \frac{\underline{s}}{\underline{e}} = \frac{1/R}{2/R + 2jC\omega + \frac{1}{jL\omega}} = \frac{1}{2+2jRC\omega+\frac{R}{jL\omega}}}$$

3/ Non-inverting amplifyer

$$\underline{H}_2 = \frac{\underline{e}}{\underline{s}} = 1 + \frac{R_2}{R_1}$$

4/ $$\underline{s}\left[2+2jRC\omega+\frac{R}{jL\omega}\right] = \underline{e} = \left(1+\frac{R_2}{R_1}\right)\underline{s}$$

$$\to 2jRC\omega\,\underline{s} + \left(2-1-\frac{R_2}{R_1}\right)\underline{s} + \frac{R}{jL\omega}\,\underline{s} = 0$$

$\times \frac{j\omega}{2RC}$

$$\to (j\omega)^2\,\underline{s} + \left(\frac{1}{2RC} - \frac{R_2}{2RC\,R_1}\right) j\omega\,\underline{s} + \frac{1}{2LC}\,\underline{s} = 0 \qquad (1)$$

$$\rightarrow \frac{d^2 s}{dt^2} + \left(\frac{1}{2RC} - \frac{R_2}{2RC\ R_1}\right)\frac{ds}{dt}$$

$$+ \frac{1}{2LC} s(t) = 0$$

Pour que les oscillations démarrent, le système doit être instable $\rightarrow \frac{1}{2RC} - \frac{R_2}{2RCR_1} < 0$

$$\rightarrow 1 < \frac{R_2}{R_1}$$

$$\rightarrow \boxed{R_2 > R_1}$$

5/ (1) $\rightarrow (j\omega)^2 + \left(\frac{1}{2RC} - \frac{R_2}{2RC\ R_1}\right) j\omega + \frac{1}{LC} = 0$

$$\rightarrow \frac{1}{LC} - \omega^2 + \frac{j\omega}{2RC}\left(1 - \frac{R_2}{R_1}\right) = 0$$

$$\rightarrow \begin{cases} \frac{1}{LC} - \omega^2 = 0 \\ 1 - \frac{R_2}{R_1} = 0 \end{cases} \rightarrow \begin{cases} \boxed{\omega = \frac{1}{\sqrt{LC}}} \\ \boxed{R_2 = R_1} \end{cases}$$

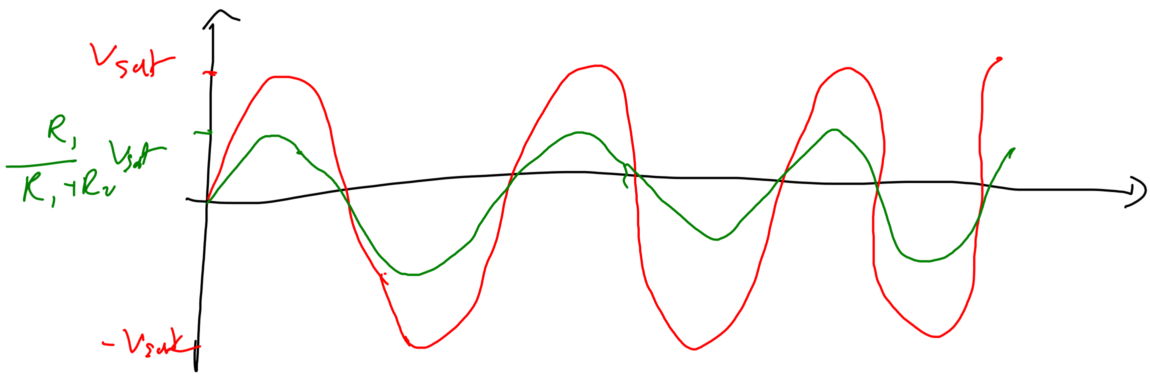
6/ L'amplitude de e est limitée par la saturation de l'ALI.

Son amplitude est donc $\underline{V_{sat}}$.

$$e = \left(1 + \frac{R_2}{R_1}\right) s$$

$$\rightarrow s = \frac{R_1}{R_1 + R_2} e$$

$\rightarrow$ L'amplitude de s est $\underline{\frac{R_1}{R_1+R_2} V_{sat}}$



7/ s est la sortie d'un filtre passe-bande donc il contient moins d'harmoniques que e

$\rightarrow$ $\boxed{\text{s est le plus sinusoïdal}}$

8/ cf 6/